Vidyalankar Institute of Technology, Mumbai | Department of Computer Engineering

Sentinel: AI-Based Intrusion Detection System with Human Review and Data Warehouse

Jai Chavan, Jeet Mane, Amaan Khan

Guided by: Dr. Kavita Shirsat | Data Warehousing and Mining

INTRODUCTION

Sentinel detects network attacks with a CNN-LSTM model. Uncertain flows go to a security analyst, and all data is stored in a warehouse for OLAP analysis.

OBJECTIVES

- Detect 12 types of attacks
- Send uncertain flows to an analyst
- Retrain only on verified feedback
- Analyse traffic using OLAP

DATASET

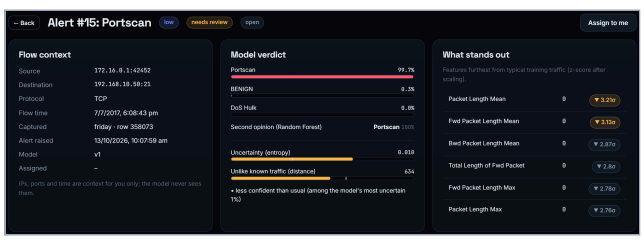
Improved CIC-IDS2017

Flows after cleaning	1,714,812
Features	64
Classes	13
Train / validation / test	70/15/15 %

Attacks: DoS (4 types), DDoS, Port scan, FTP and SSH brute force, Web attack, Botnet, Infiltration, Heartbleed.

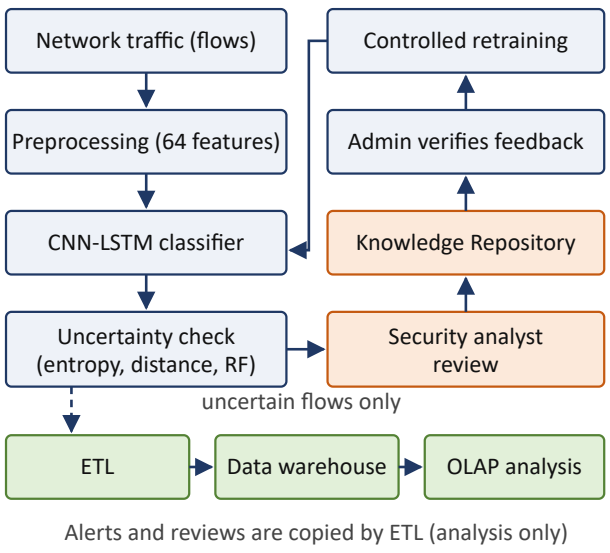
METHODOLOGY

1. Clean data, drop IPs and ports.
2. CNN-LSTM predicts the class of each flow.
3. Uncertainty check (entropy, distance, Random Forest).
4. Unsure flows go to the analyst.
5. Verified decisions retrain the model.



Analyst review screen for an uncertain flow

SYSTEM ARCHITECTURE

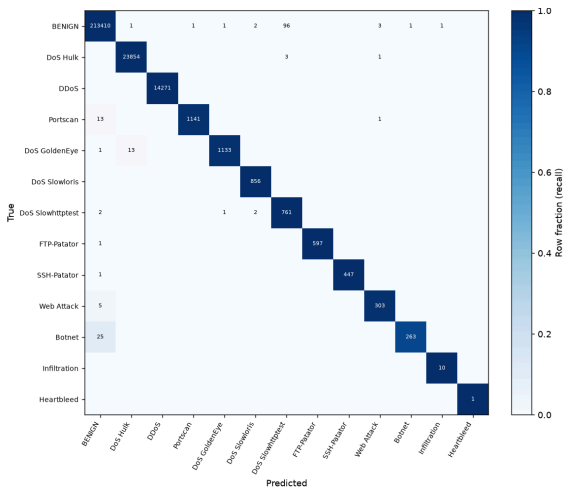


RESULTS

Model	Accuracy	Precision	Recall	F1
Random Forest	99.97	99.90	99.92	99.91
LightGBM	99.96	99.84	99.91	99.88
CNN-LSTM	**99.89**	**99.71**	**99.75**	**99.73**

Values in %, measured on the 15% test data (never used in training).

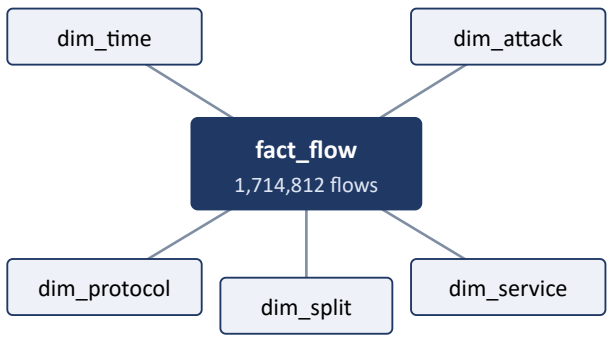
With the uncertainty check, only **1.48%** of flows go to the analyst; automatic decisions are **99.999%** correct.



Confusion matrix of the CNN-LSTM

DATA WAREHOUSE

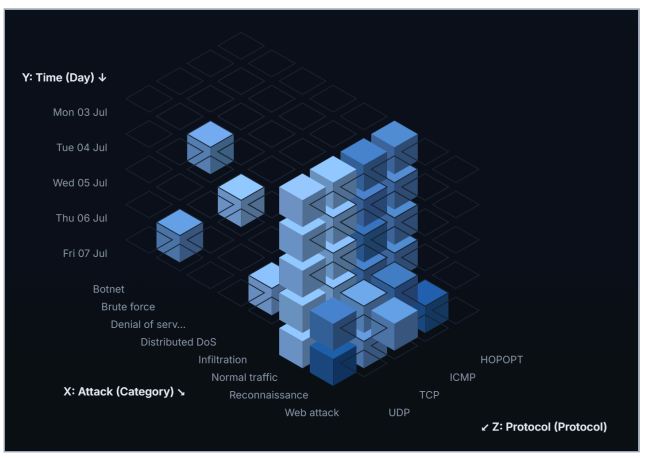
ETL (extract, transform, load) copies flows, alerts and reviews into a separate warehouse.



Star schema: one fact table and five dimensions

Also built: snowflake and fact constellation schemas, an hourly aggregate table and data marts.

OLAP ANALYSIS



OLAP cube in the web app: day x attack x protocol

- **Roll-up:** attack type to benign / malicious
- **Drill-down:** Wednesday to hours
- **Slice:** only HTTP traffic
- **Dice:** day + attack + service
- **Pivot:** swap rows and columns

Attack flows per day

Mon	0
Tue	6,972
Wed	177,517
Thu	8,148
Fri	98,732

CONCLUSION

The CNN-LSTM detects attacks with 99.73% F1, and only doubtful flows reach a human. The warehouse makes the traffic and alert history easy to analyse.

Future scope: detect slow attacks such as SSH brute force; test on CSE-CIC-IDS2018.

TOOLS USED

Python, PyTorch, scikit-learn, FastAPI, SQLite, React

Project code: github.com/JaiChavan040906/IDS